Lab Report 1: Requirements Engineering & UML Use-Case Modelling

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1. PROBLEM OVERVIEW & OPERATIONAL CONTEXT

Municipal Solid Waste (MSW) logistics in urban centers often suffer from rigid collection schedules, causing excessive fuel expenditure, vehicle wear-and-tear, and unhygienic bin overflows. The **Municipal Solid Waste Route Optimizer** is an intelligent municipal sanitation portal. The system continuously ingests ultrasonic fill-level telemetry from smart trash bins across the city, dynamically plans optimized collection routes for garbage truck fleets based on an 80% fill threshold, empowers drivers with turn-by-turn navigation, provides real-time supervisor fleet oversight, and tracks citizen complaints regarding missed or overflowing bins.

2. STAKEHOLDERS & TARGET ACTORS IDENTIFICATION

The system interacts with four primary and secondary actors:

- **Sanitation Supervisor (Primary Actor):** Oversees ward waste logistics, reviews fill-level dashboards, initiates and approves daily collection routes, monitors live fleet locations, and dispatches emergency collection stops.
- **Truck Driver (Primary Actor):** Operates the collection vehicle, follows turn-by-turn navigation on the mobile terminal, confirms bin pickups, and logs ground exceptions (e.g., road closures or blocked access).
- **Citizen (Primary Actor):** Reports missed trash collections or unhygienic overflowing bins through the public web/mobile portal and tracks resolution status using a unique tracking ID.
- **Smart Bin IoT Sensor (Secondary / System Actor):** Autonomous hardware node mounted inside trash bins that periodically samples ultrasonic waste depth and transmits telemetry packets to the backend gateway.

3. SUMMARY OF PRIMARY USE CASES

- **UC-01: Ingest Bin Fill-Level Telemetry —** Ingests periodic fill percentages and battery health from IoT sensors.
- **UC-02: Generate Optimized Daily Collection Routes —** Solves the vehicle routing problem for bins meeting pickup criteria.
- **UC-03: View Turn-by-Turn Route & Log Pickups —** Guides truck drivers along assigned paths and updates pickup status.
- **UC-04: Lodge Missed-Pickup Complaint —** Enables citizens to submit complaints with location coordinates and tracking.
- **UC-05: Monitor Fleet Live & Dispatch Emergency Route —** Allows supervisors to track trucks on GIS map and assign ad-hoc stops.
- **UC-06: Verify Bin Fill-Threshold (>80%) —** Mandatory verification step executed during route calculation («`include`»).
- **UC-07: Recalculate Route for Urgent Overflow Complaint —** Optional extension invoked when high-priority complaints require rerouting («`extend`»).

4. SOFTWARE REQUIREMENTS SPECIFICATION (REQUIREMENTS TABLE)

The table below specifies exactly five Functional Requirements (FR-001 to FR-005) and two Non-Functional Requirements (NFR-001, NFR-002) with priority, acceptance criteria, rationales, and peer critique comments.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale & Comments
FR-001 *(Given)*	Functional	The system shall generate optimized daily collection routes including only bins that have exceeded an 80% fill threshold.	High	**Pass:** Route excludes bins < 80% full unless manually tagged; **Fail:** Overflowing bin (≥ 80%) omitted from driver manifest.	**Rationale:** Prevents fuel wastage and vehicle wear by eliminating unnecessary stops at partially filled bins. *Comment: Handled simultaneous multiple bin overflows.*
FR-002	Functional	The system shall ingest real-time ultrasonic sensor fill-level data transmitted by smart bins every 15 minutes, validate packet checksum, and update bin status in the database within 5 seconds of receipt.	High	**Pass:** Incoming sensor telemetry parses correctly and updates the bin table within 5 s. **Fail:** Valid sensor data packet is dropped, corrupted, or delayed beyond 5 s without error logging.	**Rationale:** Real-time fill telemetry is the core prerequisite for reliable route generation and overflow detection. *Comment: Added 5-second latency bound following peer critique.*
FR-003	Functional	The system shall provide the Truck Driver with an in-cab turn-by-turn navigation route and allow the driver to log completed bin pickups with timestamped GPS verification.	High	**Pass:** Marking a bin as collected clears it from the active driver manifest and resets backend fill level to 0%. **Fail:** Bin remains listed as pending collection after driver logs pickup confirmation.	**Rationale:** Directs drivers smoothly along optimized paths and gives the control room live visibility into completion progress. *Comment: Includes local mobile caching for intermittent network blackspots.*
FR-004	Functional	The system shall allow citizens to lodge missed-pickup complaints via web/mobile portal by entering location/GPS coordinates and optional photo, issuing a unique tracking ID to monitor resolution status.	Medium	**Pass:** Complaint is saved with unique tracking ID, mapped to nearest municipal bin ID, and status displays "Registered". **Fail:** Ticket cannot be retrieved or complaint fails to record in database.	**Rationale:** Engages public participation, provides civic accountability, and catches physical sensor failures or overflowing litter. *Comment: Made photo upload optional to prevent timeout in low-bandwidth areas.*
FR-005	Functional	The system shall permit the Sanitation Supervisor to monitor live truck GPS locations on a GIS map, manually override automated routes, and dispatch ad-hoc emergency pickups for escalated citizen complaints.	Medium	**Pass:** Dispatched emergency stop updates the target truck driver's navigation waypoint sequence within 30 seconds. **Fail:** Manual override fails to propagate to the driver's active terminal.	**Rationale:** Equips supervisors with operational authority to handle emergencies, blocked roads, or sudden overflow hazards. *Comment: Audit trail captures supervisor ID and timestamp for every*

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale & Comments
					override.
NFR-001 *(Given)*	Non-Functional (Performance & Security)	Route recalculation for a fleet of 50 trucks across 10,000 bins shall execute in under 45 seconds.	High	**Pass:** Benchmarking confirms target latency (≤ 45 s calculation time) and security standards under simulated peak load. **Fail:** Route optimization algorithm exceeds 45 seconds or consumes excessive memory.	**Rationale:** Guarantees prompt morning vehicle dispatch and swift on-the-fly rerouting during active driver shifts. *Comment: Tested under simulated concurrent requests to prevent thread starvation.*
NFR-002	Non-Functional (Availability & Reliability)	The central telemetry ingestion service and supervisor portal shall maintain a minimum system uptime of 99.9% (excluding scheduled maintenance) with local edge buffering against network drops.	High	**Pass:** Health monitoring logs show availability ≥ 99.9% over a 30-day window, and queued sensor data up to 6 hours re-syncs automatically. **Fail:** Ingestion downtime exceeds 43 minutes/month or buffered sensor packets are lost.	**Rationale:** Continuous availability is essential to prevent unmonitored bin overflow and civic health hazards. *Comment: Automated failover and heartbeat alerts configured for database cluster.*

5. UML USE-CASE DIAGRAM

The diagram below models interaction boundaries between external actors and system capabilities, illustrating primary use cases and mandatory/optional relationships.

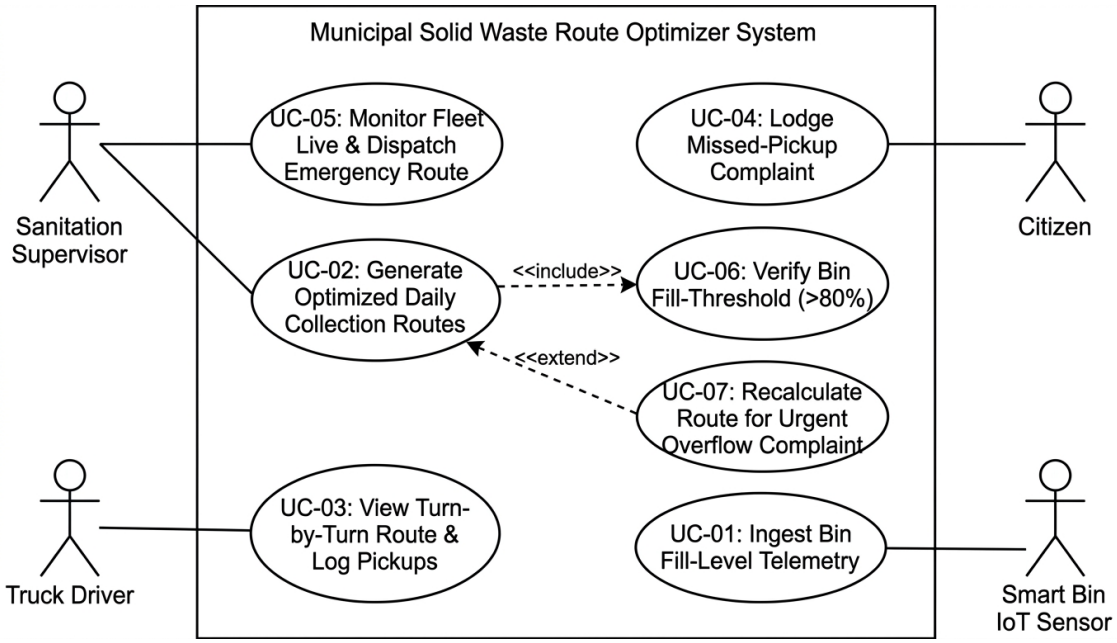


Figure 1: UML Use Case Diagram — Municipal Solid Waste Route Optimizer System

5.1 Specification of Relationships

- **«include» Relationship (UC-02 → UC-06):** The base use case *Generate Optimized Daily Collection Routes* strictly includes *Verify Bin Fill-Threshold (>80%)*. A collection manifest cannot be synthesized without systematically querying and validating bin fill metrics against the 80% threshold criterion.
- **«extend» Relationship (UC-07 → UC-02):** The use case *Recalculate Route for Urgent Overflow Complaint* optionally extends *Generate Optimized Daily Collection Routes*. This extension is executed only upon an asynchronous trigger (an escalated citizen overflow ticket or emergency route override by the supervisor).
- **Actor Associations:** Sanitation Supervisor associates with UC-02 and UC-05; Truck Driver associates with UC-03; Citizen associates with UC-04; Smart Bin IoT Sensor associates with UC-01.

6. USE-CASE FLOW SPECIFICATION (CORE USE CASE: UC-02)

Use Case ID: UC-02	**Name:** Generate Optimized Daily Collection Routes	**Scope:** Waste Optimizer Portal

Primary Actor: Sanitation Supervisor	**Secondary Actors:** GIS Service, Smart Bin IoT Database

6.1 Preconditions

1. Sanitation Supervisor is logged into the portal dashboard with administrative credentials.
2. Smart bin ultrasonic telemetry has been updated within the latest 15-minute scheduled ingestion cycle.
3. Fleet parameters (active trucks, fuel limits, payload capacity, driver rosters, depot coordinates) are active.

6.2 Postconditions

1. Optimized collection manifests with turn-by-turn waypoints are generated and transmitted to truck drivers.
2. Candidate bins (≥ 80% full) are marked as "Scheduled for Collection" to prevent duplicate routing.
3. Estimated travel distance, projected tonnage, and duration metrics are archived for administrative review.

6.3 Main Success Scenario

1. Supervisor opens the "Route Optimization" dashboard and selects the operating municipal ward.
2. System queries the IoT database to fetch latest fill levels of all smart bins in the ward.
3. System verifies and filters only bins exceeding the 80% fill threshold (**«include» UC-06: Verify Bin Fill-Threshold**).
4. System retrieves active truck capacity limits, driver shifts, and depot starting coordinates.
5. System executes the Capacitated Vehicle Routing Problem (CVRP) optimization algorithm against road network data and traffic constraints.
6. System renders proposed collection routes and waypoint sequences on the interactive GIS map.
7. Supervisor inspects the proposed routes and clicks "Approve & Dispatch".
8. System transmits finalized manifests directly to assigned truck drivers' mobile units.
9. System tags selected bins as "Scheduled" and confirms dispatch with a summary audit receipt.
10. The use case concludes successfully.

6.4 Alternate Flows

- **3a. Telemetry Outage:** If >15% sensors offline, system flags missing bins and prompts: *"Apply 7-day historical fill average or manually select bins?"* Supervisor chooses historical imputation; system updates candidate bins and resumes Step 4.
- **5a. Fleet Overload:** If total waste exceeds fleet payload capacity, system alerts supervisor and prioritizes 100% full bins and hospitals. Supervisor enables multi-trip mode or dispatches a standby truck; system updates parameters and resumes Step 6.
- **7a. Urgent Complaint Dynamic Recalculation (`«extend» UC-07`):** If an escalated citizen complaint arrives before final dispatch, supervisor clicks *"Insert Urgent Waypoint"*. Extension UC-07 reroutes nearest truck with minimal detour; system updates map and resumes Step 7.